Question Bank

Database Management Systems

Section A: CO3 — Structured Query Language (SQL)

Q.No	Question	Max. Marks	CO
1	Differentiate between Data Definition Language (DDL) and Data Manipulation Language (DML) commands in SQL. Support your answer with one example of each command type.	5	CO3
2	What are Data Control Language (DCL) and Transaction Control Language (TCL) commands? Explain GRANT, REVOKE, COMMIT, and ROLLBACK with suitable examples.	5	CO3
3	Consider the table Student(RollNo, Name, Branch, Marks, City, MOBILE NUMBER, EMAILID). Write SQL queries to: (a) count the total number of students in each branch, (b) find the average marks branch-wise, (c) find the topper's marks in the entire class, (d) find the total marks scored by each branch, (e) display the overall average marks of the class. (f) Display the names of the students whose name strats with A and end with E, having total 5 characters in name.	5	CO3
4	Explain the set operations UNION, INTERSECT, and MINUS in SQL. Illustrate each with a suitable example involving two tables.	5	CO3
5	What is a View in SQL? Explain how nested and complex queries are written, using an example involving a subquery.	5	CO3
6	Compare INNER JOIN and OUTER JOIN in SQL. Explain their working with an example involving two related tables.	5	CO3
7	Explain the use of 4 wild card operators & aggregate functions with suitable example.	5	CO3

Section B: CO4 — Relational Database Design

Q.No	Question	Max. Marks	CO
1	Define Functional Dependency. Explain the concept of closure of a set of attributes with a suitable example.	10	CO4
2	Discuss the design guidelines for a good relational schema. Why is normalization required in database design?	10	CO4
3	Explain the step-by-step process of converting a relation from 1NF to BCNF, using a suitable example at each stage.	10	CO4
4	Distinguish between transitive dependency, full functional dependency, and partial functional dependency, giving one example for each.	10	CO4
5	Explain lossy and lossless decomposition of a relation. How is lossless decomposition ensured? Illustrate with an example.	10	CO4
6	What is dependency-preserving decomposition? Explain its significance in relational database design with a suitable example.	10	CO4
7	CONVERT GIVEN RELATIONAL FUNCTIONAL DEPENDENCY INTO BCNF R(A,B,C,D,E): { A→ B, C→ D, AC→E }	10	CO4